Antimicrobial resistance is a block-nested ecosystem: economic-complexity structure and prediction of resistance emergence

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Abstract

Antimicrobial resistance (AMR) is usually monitored one pathogen–antibiotic pair at a time, ignoring the global structure of the resistance landscape. Here we represent AMR surveillance data as a bipartite pathogen–antibiotic network and analyse it with tools from Economic Complexity. Using 394 species, 43 antibiotics and two decades (2004–2024) of the ATLAS/Vivli surveillance programme, we show that the resistance network is not *globally nested* like the country–product trade network, but *block-nested* like the firm–product network: global nestedness is absent ($\mathcal{N}$ below a degree-preserving null), whereas an in-block nestedness $\mathcal{I}^*/\mathcal{N} = 7.8$ ($z = 10.3$) emerges once the network is partitioned into drug-spectrum blocks. This structure explains why the Fitness–Complexity (FC) algorithm is degenerate when applied globally, but yields meaningful rankings within blocks: antibiotic resistance complexity correlates with the WHO AWaRe classification ($\rho = 0.6$), and pathogen fitness separates ESKAPEE from non-ESKAPEE organisms ($p < 10^{-3}$), whereas the corresponding global scores fail (fitness vs. WHO priority $\rho = -0.24$, mirroring the firm–product case). Finally, projecting the bipartite network onto either layer yields a resistance-relatedness network on which the emergence of new resistance is predictable out of sample: a parameter-free network-exposure score anticipates which susceptible pathogen–antibiotic pairs become resistant in a rolling-origin test over 2020–2024 (AUC 0.71, $p < 0.005$), while resistance prevalence and proximity-to-threshold carry no information after a comparative-advantage binarization. AMR is thus a modular, locally-nested ecosystem in which resistance spreads by predictable local contagion.

1 Introduction

Antimicrobial resistance is one of the leading global health threats, associated with over a million deaths per year. Surveillance programmes such as the Pfizer ATLAS dataset (distributed through Vivli) record, for each bacterial isolate, the antibiotics it was tested against and whether it was susceptible, intermediate or resistant. These data are conventionally summarised as pairwise resistance rates: the fraction of a given species resistant to a given drug. Yet the clinical and evolutionary meaning of resistance is relational—resistance to a broadly effective, last-line drug is not equivalent to resistance to an already-compromised one, and the drugs a pathogen resists are not independent draws but reflect shared mechanisms and co-selection.

The Economic Complexity (EC) framework was developed to extract exactly this kind of relational information from bipartite networks [1, 2]. Applied to the country–product export network, it uncovers a nested structure that underlies the Fitness–Complexity (FC) algorithm and enables data-driven forecasts of economic growth. The same toolbox—bipartite projections, relatedness networks, fitness and complexity—has since been applied to jobs and skills [3], to chess openings [4], and, crucially for us, to firms [5], where the network was found to be *block-nested* rather than globally nested, so that FC must be applied *within* locally-nested blocks.

Here we ask whether the AMR pathogen–antibiotic network can be understood through the same lens, and whether its structure makes resistance emergence predictable. We find that AMR behaves like the firm ecosystem: it is block-nested, its blocks correspond to antibiotic spectrum, within-block FC scores agree with independent WHO expert classifications, and a network-exposure predictor forecasts new resistance out of sample. We report the results first, then discuss implications and methods.

2 Results

2.1 From resistance rates to a resistance network

The starting point is a simple table: for every bacterial species and every antibiotic, what fraction of the tested samples were resistant. We build this table from two decades of surveillance (394 species, 43 antibiotics, 2004–2024), keeping only well-sampled cases with at least 30 tested isolates (11 507 such cases). The raw resistance fraction r_{pa} (Fig. 1a) is, however, a misleading quantity to compare across the table, because it is dominated by two trivial effects: some drugs (ampicillin, penicillin) are old and beaten by almost everything, and some species resist almost everything. A high number can therefore mean either a genuinely notable resistance or simply that the drug is generally weak.

To strip out these size effects we use a measure borrowed from economics, the *revealed comparative advantage* (RCA), which asks a sharper question: does this species resist this drug *more than one would expect by chance*, given how resistant the species is overall and how vulnerable the drug is overall? When it does—$\text{RCA}_{pa} \geq 1$—we mark an effective resistance link (Fig. 1b). This cutoff is not tuned for our results: on our data RCA = 1 falls right at the *median* of the RCA distribution (median RCA = 0.99–1.17 across time windows), so it simply separates "more resistant than typical" from "less". The outcome is a clean yes/no matrix M_{pa} (Fig. 1c), and this is the object we analyse throughout. The step matters: as we show below, it removes exactly the confounds—how common a resistance is, and how close a species already sits to the threshold—that would otherwise masquerade as prediction, leaving only the genuine network signal.

2.2 A modular landscape, not a single ladder

The first question is how the resistance table is organised as a whole. In world trade, the analogous country–product table is *nested* like a ladder: the most developed countries export nearly everything, less developed ones export only a subset of the same common goods, and the rare, sophisticated products are made only at the top. If AMR worked the same way, there would be a single ladder of "super-resistant" bugs at the top that beat every drug, down to specialists at the bottom. We test this with a standard nestedness score $\mathcal{N}$ and its within-group version $\mathcal{I}$ [6, 5].

AMR does *not* form a single ladder. Measured globally, its nestedness is negligible ($\mathcal{N} = 0.05$)—in fact slightly *below* what one would get by chance ($z = -4.3$). But this is not disorder: when we first split the drugs and bugs into groups (by spectral co-clustering) and measure nestedness *inside* each group, it jumps to $\mathcal{I}^* = 0.39$, almost eight times the global value ($\mathcal{I}^*/\mathcal{N} = 7.8$; far above chance, $z = 10.3$; Fig. 2b). In other words, the resistance landscape is *modular*: it is made of several separate ladders rather than one. This is exactly the pattern Ref. [5] found for companies (ratio

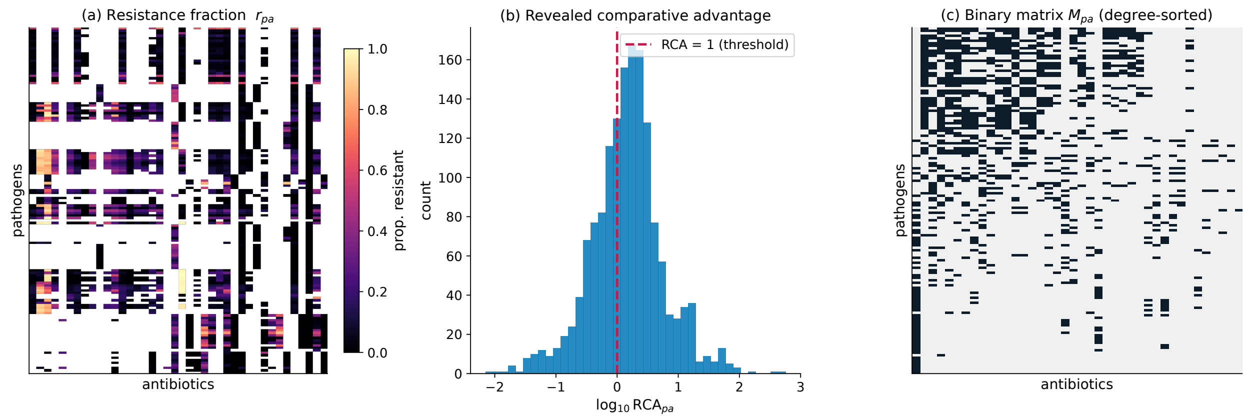

Figure 1: Construction of the bipartite resistance network. (a) Resistance fraction r_{pa} for pathogens (rows) × antibiotics (columns). (b) Distribution of the revealed comparative advantage RCA_{pa}; the threshold $\mathrm{RCA} = 1$ (dashed) defines effective resistance links. (c) Resulting binary matrix M_{pa}, degree-sorted.

$\approx$ 12) and the opposite of the one for countries ($\approx$ 1.02)—AMR sits firmly on the company-like, modular side.

Reassuringly, the groups the algorithm finds are the ones a microbiologist would draw (Fig. 2a): a large Gram-negative β-lactam block (Enterobacterales), a Gram-positive block, a block of enterococci resisting glycopeptides (the VRE ecosystem), and an anaerobe block. So the ladder structure is real but *local*: within a single therapeutic ecosystem the bugs that beat the rare drugs also beat the common ones, but this ordering simply does not carry across ecosystems—a drug that is hard to resist for a Gram-negative says nothing about a Gram-positive.

2.3 Ranking bugs and drugs, and checking it against the WHO

Economic Complexity summarises such a table with two intuitive scores. A country's *fitness* is high if it exports many things, especially hard ones; a product's *complexity* is high if only the most capable countries make it. We compute the same two scores here [2], reading a pathogen as a "country" that carries resistances and a drug as a "product": a bug's fitness is then the breadth and sophistication of its resistance arsenal, and a drug's complexity is high when only the most resistant bugs manage to beat it—the mark of a last-line agent. The obvious test is whether these purely data-driven scores agree with what clinicians already know.

Computed on the whole table at once, the scores are useless. The modular structure of the previous section breaks the algorithm, which assumes a single ladder: one group (enterococci resisting vancomycin) monopolises the top, and the ranking ends up *anti*-correlated with clinical threat (global pathogen fitness vs. the WHO priority list, $\rho = -0.24$; Fig. 3b). This is not a bug in our data but the same failure Ref. [5] reported for companies, and it is the practical reason the algorithm must be run inside the blocks rather than across them.

Computed *inside each block*, the same scores line up with independent expert judgement. Drug complexity tracks the WHO AWaRe classification, which ranks antibiotics from freely-used "Access" to protected last-resort "Reserve" (Spearman $\rho = 0.47$ overall, 0.59 in-block, 0.62 in the Gram-negative block, $p \leq 10^{-3}$; Fig. 3a): the most "complex" resistances are precisely those to Reserve agents (cefiderocol, ceftazidime–avibactam, carbapenems, linezolid). Likewise, within-block bug fitness cleanly separates the ESKAPEE pathogens (the ESKAPE priority group plus *E. coli*)—which cause most hospital resistance—from the rest (median fitness +1.0 vs. −0.4; Mann–Whitney

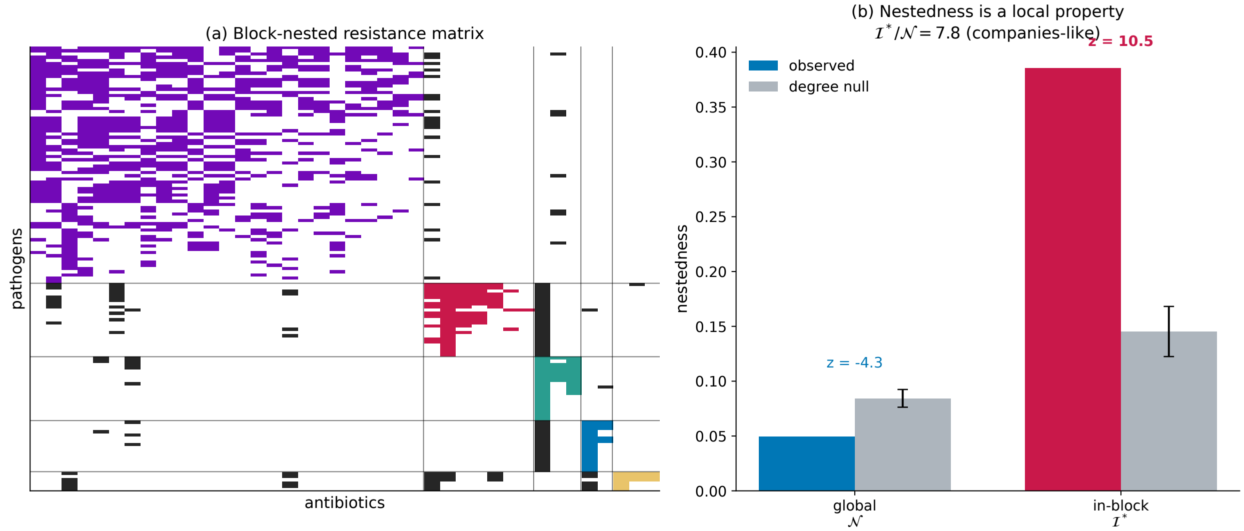

Figure 2: AMR is block-nested. (a) The binary resistance matrix reordered by co-clustered blocks (colours) and by degree within each block; off-block links in dark grey. Each block is internally nested (triangular). (b) Global nestedness $\mathcal{N}$ and in-block nestedness $\mathcal{I}^*$ versus a degree-preserving null. $\mathcal{N}$ is below the null ($z = -4.3$) while $\mathcal{I}^*$ is far above it ($z = 10.3$), giving $\mathcal{I}^*/\mathcal{N} = 7.8$.

$p < 10^{-3}$; Fig. 3c), with *Klebsiella pneumoniae* and *Enterobacter cloacae* at the top. The sign literally flips between the global and the in-block view (Fig. 3b vs. c), and that flip is the cleanest confirmation of the modular picture: only the local scores agree with the WHO.

2.4 Predicting where resistance appears next

The most useful question is whether this structure lets us *anticipate* new resistance. The idea is that resistances are not independent: drugs that share a mechanism tend to be beaten together. We make this concrete by building a *co-resistance network* of drugs, linking two antibiotics when many bugs resist both (Fig. 4a)—its links duly organise by drug family and AWaRe class. For a species not yet resistant to a drug a, we then ask a single question: of the drugs that sit next to a in this network, how many does the species already beat? This fraction, the network-exposure score ρ, is the same "adjacent-possible" logic used to forecast which chess openings a player will adopt next [4]. The bet is that a species surrounded by drugs it already resists is one short step from resisting a too.

We stress that this predictor has *no free parameters to tune*: both the yes/no cutoff (RCA ≥ 1, the median) and the way the network is built are fixed in advance, so there is nothing that could be quietly adjusted to flatter the result. This lets us test it in the fairest possible way—strictly forward in time. For each year we build everything from the past only and predict the following year's new resistances, repeating this for every year from 2019 to 2023 (predicting 2020–2024); each prediction concerns a future the model has never seen. The score works, and works consistently. Bugs in the most-exposed group are more than *three times* more likely to acquire the new resistance than those in the least-exposed group (17% vs. 4.6%; Fig. 4b). Its overall accuracy, measured as the probability of ranking a true future resistance above a non-event (the AUC), is 0.71 for the drug network and 0.63 for the analogous pathogen network (two species linked when they resist similar drugs), stable across every year tested (0.67–0.77 and 0.59–0.69). To be sure this reflects the real wiring and not just how busy each node is, we reshuffled the network labels 200 times: the

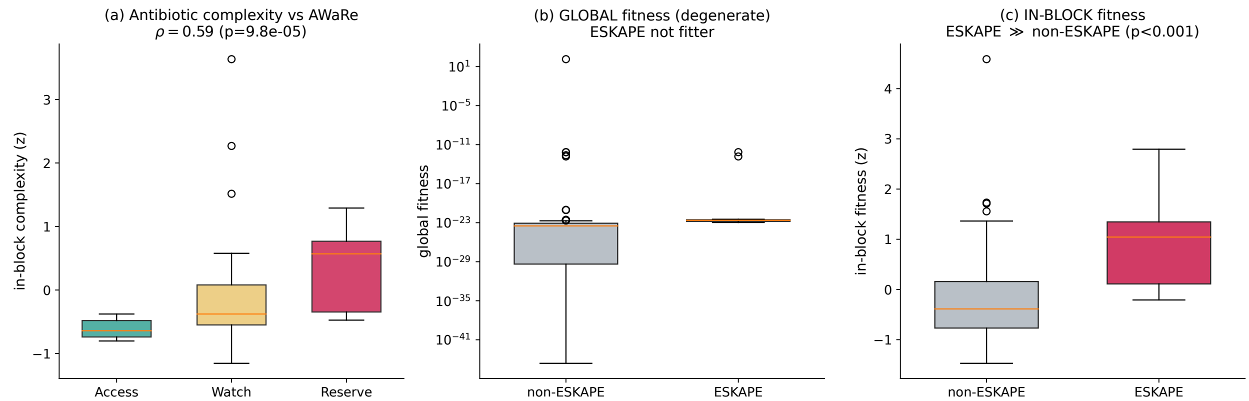

Figure 3: Fitness and complexity validate against WHO classifications. (a) In-block antibiotic complexity increases with the WHO AWaRe tier (Access/Watch/Reserve), Spearman $\rho = 0.6$. (b) *Global* pathogen fitness fails: ESKAPEE organisms are not the fittest (fitness vs. WHO priority $\rho = -0.24$). (c) *In-block* pathogen fitness succeeds: ESKAPEE organisms are markedly fitter than non-ESKAPEE ($p < 10^{-3}$).

true score stands far above this chance baseline ($z = 6.2$ and 8.7, $p < 0.005$).

The comparison that makes the point is with the obvious naïve guesses. Once resistance is measured by comparative advantage, simply betting on the currently most common resistances, or on the species that already sit closest to the threshold, predicts nothing at all (AUC ≈ 0.51, i.e. a coin flip; Fig. 4c). Only the network position carries information. Finally, reading the score drug by drug flags a shortlist of “contagion-hub” antibiotics—ceftazidime, meropenem, gentamicin, cefepime (Fig. 4d)—whose emerging resistance is the most anticipatable from a species’ existing profile, and which are therefore the natural targets for early warning.

This local-contagion picture can be read directly off the relatedness networks. Fig. 5 follows worked examples over two decades: two pathogens (*Klebsiella pneumoniae*, *Escherichia coli*) progressively light up the antibiotic co-resistance network as they acquire resistances, and two antibiotics (meropenem, ciprofloxacin) spread across the pathogen relatedness network as more species become resistant. In every case the newly-lit nodes are network neighbours of already-lit ones—resistance diffuses through the network rather than appearing at random—which is exactly the regularity the exposure score exploits.

3 Discussion

Representing AMR surveillance as a bipartite network and analysing it with Economic Complexity tools yields a coherent picture that departs from the trade paradigm. AMR is *not* a global complexity ladder: there is no single axis along which pathogens diversify their resistance and drugs become progressively harder to overcome. Instead, the network is block-nested—modular by antibiotic spectrum, with nestedness re-emerging inside each therapeutic ecosystem. This is the same structure found for firms rather than countries [5], and it has the same methodological consequence: the Fitness–Complexity algorithm, which presupposes global nestedness, is degenerate when applied to the whole matrix and must instead be run within the locally-nested blocks. The strong external validation of the within-block scores against the WHO AWaRe and priority-pathogen classifications—together with the failure of the global scores—turns this from a technical caveat into a positive, testable finding.

The practical payoff is prediction. Once resistance is binarized by comparative advantage, neither how common a resistance is nor how close a pathogen already sits predicts which resistances emerge next; only network position does. Resistance spreads in the "adjacent possible", appearing where a pathogen is already close, on the co-resistance network, to what it resists—mirroring how chess players adopt openings adjacent to their repertoire [4]. The two projections give complementary but consistent views, and the per-drug decomposition provides an actionable early-warning list of antibiotics whose emerging resistance is most anticipatable.

Several limitations temper these conclusions. The predictive AUCs are moderate and rest on a limited number of acquisition events; genus/species coverage is uneven across the surveillance programme; and RCA-on-resistance, while natural, is a modelling choice. The blocks are validated against a degree-preserving null and against expert classifications, but a fully Bayesian bipartite-configuration treatment and multi-year prediction labels would sharpen the estimates. Nonetheless, the convergence of structure, validated rankings and out-of-sample prediction suggests that the economic-complexity view of AMR is both interpretable and useful, and that surveillance data contain exploitable early signals of where resistance will emerge next.

4 Methods

Data. We use the ATLAS/Vivli antimicrobial susceptibility surveillance programme, aggregated to species $\times$ antibiotic $\times$ year counts of susceptible/intermediate/resistant isolates. We retain cells with $N_{\text{tested}} \geq 30$. Unless noted, structural and validation analyses use the pooled 2015–2024 window (139 pathogens $\times$ 40 antibiotics); prediction uses expanding causal windows (below).

Binarization. From the pooled resistance fraction $r_{pa} = n^R_{pa}/N_{pa}$ we compute the revealed comparative advantage $\mathrm{RCA}_{pa} = \big(r_{pa}/\sum_{a'} r_{pa'}\big)/\big(\sum_{p'} r_{p'a}/\sum_{p'a'} r_{p'a'}\big)$ and set $M_{pa} = 1$ iff $\mathrm{RCA}_{pa} \geq 1$, following the standard EC threshold [5]. On these data this threshold coincides with the median of the RCA distribution (median RCA $\in [0.99, 1.17]$ across windows), so the binarization is fixed a priori and is not tuned against any outcome.

Nestedness and in-block nestedness. We use the overlap-based global nestedness $\mathcal{N}$ and in-block nestedness $\mathcal{I}$ of Refs. [6, 5]. For row nodes, $\mathcal{I} \propto \sum_{i,j} \frac{O_{ij} - \langle O_{ij}\rangle}{k_j(C_i - 1)}\, \Theta(k_i - k_j)\, \delta(a_i, a_j)$ with an analogous column term, where O_{ij} is the number of shared links, $\langle O_{ij}\rangle = k_i k_j / N$ the degree-null expectation, C_i the size of i's block and Θ the Heaviside function; $\mathcal{N}$ is the single-block case. Blocks are obtained by spectral co-clustering (we optimise over $k = 2-6$). Significance is assessed against 100 degree-preserving randomisations.

Fitness–Complexity. We iterate the FC map [2] $\tilde{F}^{(n)}_p = \sum_a M_{pa} Q^{(n-1)}_a$, $\tilde{Q}^{(n)}_a = 1/\sum_p M_{pa}/F^{(n-1)}_p$, normalising each iterate by its mean, to convergence. Global scores use the full matrix; in-block scores are computed on each block's sub-matrix and z-scored within block before pooling.

Projections. One-mode projections use the plain co-occurrence weighting $W_{ab} = \sum_p M_{pa} M_{pb}$—the number of species resistant to both drugs a and b, symmetric by construction—for the antibiotic layer, and the analogous $W_{pq} = \sum_a M_{pa} M_{qa}$ for pathogens. We also verified that a bipartite-configuration (BiCM) validated projection yields essentially no edges, consistent with co-resistance being degree-explained. Among the projection weightings we tried, this raw co-occurrence gave the best out-of-sample prediction: degree-normalised weightings consistently *lowered* the AUC (e.g. $0.71 \to 0.67$ under a symmetric degree normalisation), because the common, widely-shared co-resistances they down-weight are exactly what carries the predictive signal here.

Network-exposure prediction. The predictor is deliberately parameter-free, so that no quantity is tuned against the test set: the binarization (RCA ≥ 1, the distribution median) and the projection (plain co-occurrence) are both fixed a priori, and the exposure score is a pure ranking. For each origin year t we build M, the projection W and each pathogen's repertoire from data with year $\leq t$, and define the exposure $\rho_{pa} = \sum_o W_{ao} M_{po} / \sum_o W_{ao}$. Candidates are (p, a) pairs susceptible and observed at t and observed at $t+1$; the label is RCA-resistance at $t+1$. Because there is nothing to fit, we evaluate by *rolling origin*: every origin $t = 2019 \ldots 2023$ (predicting $2020 \ldots 2024$) is a genuine out-of-sample test, and we report the rank-based AUC per origin and pooled. Significance uses 200 permutations of the network node labels. Baselines are resistance prevalence and current proximity-to-threshold. A separate robustness sweep over alternative binarizations (absolute thresholds, RCA ≥ 1.5) and projection weightings (degree-normalised, BiCM-validated) confirms that the plain co-occurrence choice is the simplest available and is not optimised against the test set—if anything the alternatives we tried predicted less well.

External validation. Antibiotic complexity is compared (Spearman) to the WHO AWaRe class (Access/Watch/Reserve), taken from the WHO 2023 AWaRe classification [7] (all 43 antibiotics mapped; see the reference table shipped with the data). Pathogen fitness is compared to the WHO bacterial-priority tier and to ESKAPEE membership (the ESKAPE priority pathogens plus *E. coli*; Mann–Whitney).

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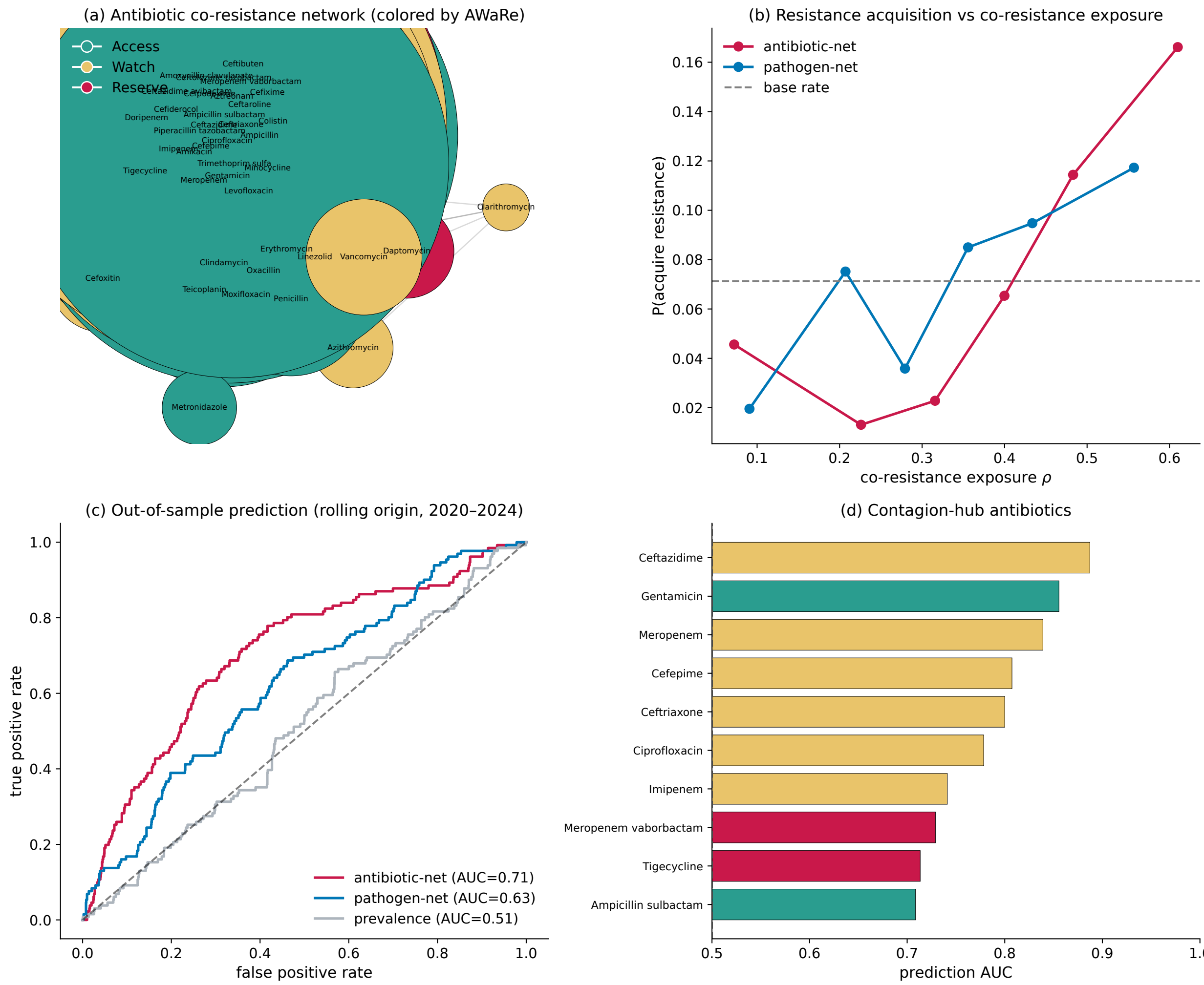

Figure 4: Prediction of resistance emergence from network exposure. (a) Antibiotic co-resistance network (co-occurrence projection), nodes coloured by WHO AWaRe class, sized by degree. (b) Probability of acquiring a new resistance versus network-exposure ρ for the antibiotic and pathogen projections; dashed line is the base rate. (c) Out-of-sample ROC (pooled rolling-origin test): both network scores beat the prevalence baseline, which is uninformative under RCA. (d) Contagion-hub antibiotics: per-drug prediction AUC (colour = AWaRe class).

Figure 5: Spreading of resistance across the relatedness networks. Top two rows: two example pathogens (*K. pneumoniae*, *E. coli*) shown on the fixed antibiotic co-resistance network; a node is coloured when the pathogen has acquired RCA-resistance to that drug by the indicated year (cumulative, so nodes light up and never revert). Bottom two rows: two example antibiotics (meropenem, ciprofloxacin) shown on the pathogen relatedness network; a node is coloured when that species has become resistant. Resistance spreads to network-adjacent nodes rather than at random.